Differences between Blotting Techniques

	Southern blotting	Northern blotting	Western blotting
Name	Southern name of the inventor	Northern a misnomer	Western a misnomer
Separation of	DNA	RNA	Proteins
Denaturation	Needed	Not needed	Needed
Membrane	Nitrocellulose/ nylon	Amino benzyloxymethyl	Nitrocellulose
Hybridisaiton	DNA-DNA	RNA-DNA	Protein-antibody
Visualising	Autoradiogram	Autoradiogram	Dark room

Table 4.2: Difference between Blotting Techniques

2 The replacement vector has the homology region and it is co-linear with target. This vector is linearized prior to transfection outside the homology region and then consequently a crossing over occurs to replace the endogenous DNA with the incoming DNA.

Transfection: Introduction of foreign nucleic acids into cells by non-viral methods.

4.7.7. Genome Sequencing and Plant Genome Projects

The whole complement of genes that determine all characteristics of an organism is called genome. Which may be nuclear genome, mitochondrial genome or plastid genome. Genome of many plants contain both functional and non-expressive DNA proteins. Genome project refers to a project in which the whole genome of plant is analysed using sequence analysis and sequence homology with other plants. Such genome projects have so far been undertaken in *Chlamydomonas*(algae), *Arabidopsis thaliana*, rice and maize plants.

Genome content of an organism is expressed in terms of number of base pairs or in terms of the content of DNA which is expressed as c-value.

4.7.8 Evolutionary pattern assessed using DNA.

In recent years the evolutionary relationship between different plant taxa is assessed using DNA content as well as the similarities and differences in the DNA sequence (sequence homology). Based on such analysis the taxa and their relationship are indicated in cladogram. Which will show the genetic distance between two taxa. It also shows antiquity or modernity of any taxon with respect to one another (See also Unit-2, Chapter-5 of XI Std.)



Barcode: You might have seen in all books barcoding and also in items you buy in supermarket. This will reveal the identity of the book or item as well the details like prize. Similarily, Barcode in genetic term refer to the identify of the taxon based on its genetic makeup. In practice, it is an optical, machine-readable representation of data which describes about the characters of any plants or any objects.



4.7.9 Genome editing and CRISPR - Cas9

Genome editing or gene editing is a group of technologies that has the ability to change an organism's DNA. These technologies allow genetic material to be added, removed, or

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